EXP 9: Compare Linear and Polynomial Regression using Python

CODING:

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# Linear and Polynomial Regression

X = [1, 2, 3, 4, 5, 6]

Y = [1, 4, 9, 16, 25, 36]

n = len(X)

# Linear Regression

mean_x = sum(X) / n

mean_y = sum(Y) / n

numerator = 0

denominator = 0

for i in range(n):

   numerator += (X[i] - mean_x) * (Y[i] - mean_y)

   denominator += (X[i] - mean_x) ** 2

slope = numerator / denominator

intercept = mean_y - slope * mean_x

linear_pred = []

for x in X:

   linear_pred.append(slope * x + intercept)
```

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# Polynomial Regression (Degree 2)
poly_pred = []

for x in X:
    y = x ** 2
    poly_pred.append(y)

# Calculate Mean Squared Error
linear_error = 0
poly_error = 0

for i in range(n):
    linear_error += (Y[i] - linear_pred[i]) ** 2
    poly_error += (Y[i] - poly_pred[i]) ** 2

linear_mse = linear_error / n
poly_mse = poly_error / n

# Display Results
print("Actual Values:", Y)

print("\nLinear Regression:")
print("Predicted Values:", linear_pred)
print("Mean Squared Error:", round(linear_mse, 2))

print("\nPolynomial Regression:")
print("Predicted Values:", poly_pred)
```

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print("Mean Squared Error:", round(poly_mse, 2))


print("\nComparison:")
if linear_mse < poly_mse:
    print("Linear Regression performs better.")
else:
    print("Polynomial Regression performs better.")
```

OUTPUT:

